

AMRA IBRAHIM

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EDUCATION

Virginia Commonwealth University

Bachelor of Science in Computer Science (Software Engineering), Minor: Artificial Intelligence

Richmond, VA
Expected May 2027

Security Clearance: Department of Defense - Secret Clearance (Active)

Relevant Coursework: Computer Systems, Theory of Computation, Databases, Programming Languages, Data Structures and Algorithms, Object-Oriented, Calculus I- II, Linear Algebra, Discrete Structures, Data Ethics

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C/C++, JavaScript, SQL, HTML/CSS, React.js, Node.js, Express.js

Libraries & Tools: Pandas, NumPy, Scikit-learn, PyTorch, LangChain, API Integration, Lens Studio, MLflow

Developer Tools & Platforms: Git, AWS, Linux, Jira, Agile, IntelliJ, Visual Studio, Jupyter Notebook, PyCharm, REST APIs

PROFESSIONAL EXPERIENCE

LinkedIn

Software Engineer Intern

San Francisco, CA
May 2026-Aug 2026

- Engineered an end-to-end Azure developer sentiment (CSAT/DSAT) survey system in, integrating four internal Azure tooling and telemetry sources through a ports-and-adapters architecture with independent source failure isolation.
- Designed and implemented a secure, read-only data access layer across externally managed databases with strict domain separation, alongside an owned MySQL datastore with Alembic-managed schema migrations for survey persistence.
- Developed and deployed a Slack Block Kit survey bot using outbound Socket Mode with no inbound ingress, incorporating survey-fatigue controls, including a 7-day lookback, 7-day cooldown, monthly cap, and non-production safety guardrails, with end-to-end validation.

Leidos

AI/ML Software Engineer Intern: SMIT (Navy-Marine Corps Intranet)

Reston, VA
Feb 2026-May 2026

- Built and refined multi-agent AI orchestration workflows for the Navy's SMIT ITSM portfolio, automating summarization and analysis of service-desk ticket, network sensor, and performance data to accelerate operational decision-making.
- Prototyped applied-AI insight pipelines in Python and JavaScript using vector databases and enterprise API integrations (Jira, Splunk, ServiceNow, Ask Sage, Azure Data) to enable trend analysis, forecasting, and anomaly detection across enterprise IT operations.
- Designed prompt architectures and agent workflows, and built lightweight Flask/React UI components to visualize AI-generated insights; delivered working software in Agile ceremonies to a geographically distributed team.

Dell Technologies

Software Engineer Intern

Round Rock, TX
May 2025-Aug 2025

- Developed an AI/ML pipeline with Retrieval-Augmented Generation (RAG) and Kafka integration to analyze Dell server telemetry and support logs in real time, enabling instant diagnostic insights and resolution suggestions for engineers.
- Designed automated ETL workflows using Python, Pandas, and vector databases (FAISS/ChromaDB) to clean and process 14K+ enterprise log and ticket entries, improving data quality and retrieval accuracy while reducing noise and redundancy.
- Optimized pipelines for scalability on NVIDIA GPU infrastructure (GB200-ready), cutting inference latency by 75% and demonstrating how AI-powered log analysis can enhance support efficiency, reduce triaging time, and improve customer experience.

TECHNICAL PROJECTS

JP Morgan Chase – Data for Good Hackathon | 1st Place (Machine Learning & Social Impact)

- Developed and deployed a logistic regression model in Python (Pandas, NumPy, Scikit-learn) to predict employment outcomes for domestic violence survivors based on certification, education, and financial indicators.
- Performed feature engineering and exploratory analysis on program data to identify barriers to workforce entry, uncovering that certifications alone did not significantly increase employability without additional career support.
- Partnered with Of Home, Family & Future (OHFF) to create a data-backed proposal for a youth career readiness program, using insights to recommend early intervention for children of survivors; presented findings to JPMorgan judges and won 1st place.

Local Drink Finder | AI-powered café and boba drink matcher

- Administered an AI-powered drink finder app that helps users locate nearby cafés based on natural language drink requests and ZIP code input.
- Implemented semantic search with sentence-transformers and LLMs (Mistral/Hugging Face) to match user queries with generated café menus, filtering out irrelevant results.
- Deployed both local and cloud-based versions using Google Maps API, enabling real-time recommendations and improving search relevance for underrepresented or culturally specific drinks.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

ColorStack Chapter @ VCU | Founder & President | Richmond, VA

August 2025-Present

- Founded the first ColorStack chapter at VCU, building an inclusive community that supports underrepresented students in Computer Science.
- Led partnerships, events, and executive board operations to deliver career development, technical, and networking opportunities for members.

Management Leadership for Tomorrow | Career Preparation Fellow | Washington, DC (Virtual)

January 2025-Present

- Accepted into an intensive 18-month personal and professional development program for high achieving diverse talent.
- Engaged in business case studies and attended industry leader conferences, enhancing my understanding of Data Analytics and Cloud Technologies, and Software Engineering.